

MATH 211: CALCULUS III

Spring 2018–19 • Second Exam • 15 April 2019

Q1. [5 pts each] *In each part, choose the correct answer.*

i. The volume of the solid that lies below the plane $2x - y + z = 5$ and above the square $[0, 2] \times [1, 3]$ is _____.

- (a) 5 (b) 10 (c) 16 (d) 18 (e) 20

ii. The volume of the solid enclosed by the cylinder $x^2 + y^2 = 1$ and the planes $z = 0$ and $z = 2 + x$ is _____.

- (a) $\frac{\pi}{2}$ (b) π (c) 2π (d) $\frac{5\pi}{6}$ (e) $\frac{3\pi}{2}$

iii. Which of the following is a representation in cylindrical coordinates of the point $(x, y, z) = (1, -1, 1)$?

- (a) $(\sqrt{2}, \frac{7\pi}{4}, 1)$ (b) $(\sqrt{3}, -\frac{\pi}{4}, 1)$ (c) $(-\sqrt{2}, \frac{5\pi}{4}, 1)$
(d) $(\sqrt{2}, -\frac{\pi}{4}, 3)$ (e) $(\sqrt{2}, \frac{3\pi}{4}, 1)$

iv. Which of the following is the equation in spherical coordinates of the surface $z = \sqrt{3x^2 + 3y^2}$?

- (a) $z = \sqrt{3}r$ (b) $\phi = \pi/6$ (c) $\theta = \pi/6$
(d) $\phi = \pi/3$ (e) $\theta = \pi/3$

v. $\int_{-2}^2 \int_0^{\sqrt{4-x^2}} \int_0^{4-x^2-y^2} (3x^2 + 3y^2) dz dy dx = \underline{\hspace{2cm}}.$

- (a) -8π (b) 8π (c) 26π (d) 16π (e) 32π

Q2. [15 pts] Evaluate $\int_0^4 \int_{\sqrt{y}}^2 e^{(x^3)} dx dy$ by reversing the order of integration.

Q3. [15 pts] Find the area of the part of the surface $z = x^2 + 3$ that lies above the triangular region in the xy -plane with vertices $(0, 0)$, $(1, 2)$, and $(1, -1)$.

Q4. [15 pts] Use spherical coordinates to evaluate $\iiint_E z(x^2 + y^2 + z^2) dV$ where E is the ball $x^2 + y^2 + z^2 \leq z$.

Q5. [15 pts] Let T be the solid tetrahedron with vertices $(0, 0, 0)$, $(2, 0, 0)$, $(0, 2, 0)$, and $(0, 0, 2)$. Find the mass of T if the density at each point of T is $\rho(x, y, z) = y^2$.

Q6. [15 pts] Let R be the region in the xy -plane given by the inequality $|x| + |y| \leq 1$. Use an appropriate change of variables to evaluate $\iint_R e^{y-x} dA$.

Q1. i. The correct choice is 20.

$$V = \int_1^3 \int_0^2 (5 - 2x + y) \, dx \, dy = \int_1^3 \left[(5 + y)x - x^2 \right]_0^2 \, dy = \int_1^3 (6 + 2y) \, dy = \left[6y + y^2 \right]_1^3 = 20.$$

ii. The correct choice is 2π .

$$V = \int_0^{2\pi} \int_0^1 (2 + r \cos \theta) r \, dr \, d\theta = \int_0^{2\pi} \left[r^2 + \frac{1}{3} r^3 \cos \theta \right]_0^1 \, d\theta = \int_0^{2\pi} \left(1 + \frac{1}{3} \cos \theta \right) \, d\theta = \left[\theta + \frac{1}{3} \sin \theta \right]_0^{2\pi} = 2\pi.$$

iii. The correct choice is $(\sqrt{2}, \frac{7\pi}{4}, 1)$.

iv. The correct choice is $\phi = \pi/6$.

v. The correct choice is 16π .

$$I = \int_0^\pi \int_0^2 \int_0^{4-r^2} 3r^2 r \, dz \, dr \, d\theta = 3\pi \int_0^2 r^3 (4 - r^2) \, dr = 3\pi \left[r^4 - \frac{1}{6} r^6 \right]_0^2 = 3\pi \times 16 \left(1 - \frac{2}{3} \right) = 16\pi.$$

Q2. Let $D = \{(x, y) \mid 0 \leq y \leq 4, \sqrt{y} \leq x \leq 2\}$. Then

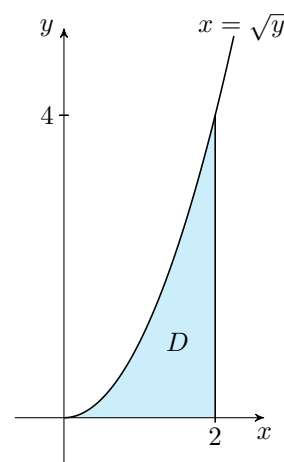
$$I = \int_0^4 \int_{\sqrt{y}}^2 e^{x^3} \, dx \, dy = \iint_D e^{x^3} \, dA.$$

On the other hand, using the following sketch of D we see that

$$D = \{(x, y) \mid 0 \leq x \leq 2, 0 \leq y \leq x^2\},$$

hence

$$I = \int_0^2 \int_0^{x^2} e^{x^3} \, dy \, dx = \frac{1}{3} \int_0^2 3x^2 e^{x^3} \, dx = \frac{1}{3} \left[e^{x^3} \right]_0^2 = \frac{e^8 - 1}{3}.$$



Q3. Let D denote the given triangular region. Then

$$D = \{(x, y) \mid 0 \leq x \leq 1, -x \leq y \leq 2x\},$$

and we have

$$\begin{aligned} A(S) &= \iint_D \sqrt{1 + \left(\frac{\partial z}{\partial x} \right)^2 + \left(\frac{\partial z}{\partial y} \right)^2} \, dA = \iint_D \sqrt{1 + 4x^2} \, dA \\ &= \int_0^2 \int_{-x}^{2x} \sqrt{1 + 4x^2} \, dy \, dx = \int_0^2 3x \sqrt{1 + 4x^2} \, dx = \frac{3}{8} \cdot \frac{2}{3} \left[(1 + 4x^2)^{3/2} \right]_0^1 = \frac{5\sqrt{5} - 1}{4}. \end{aligned}$$

Q4. The solid E can be described in spherical coordinates by

$$E = \{(\rho, \theta, \phi) \mid 0 \leq \theta \leq 2\pi, 0 \leq \phi \leq \frac{\pi}{2}, 0 \leq \rho \leq \cos \phi\},$$

using which we obtain

$$\begin{aligned} \iiint_E z(x^2 + y^2 + z^2) \, dV &= \int_0^{2\pi} \int_0^{\pi/2} \int_0^{\cos \phi} \rho \cos \phi \cdot \rho^2 \cdot \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta \\ &= 2\pi \int_0^{\pi/2} \left[\frac{1}{6} \rho^6 \right]_0^{\cos \phi} \cos \phi \sin \phi \, d\phi \\ &= \frac{\pi}{3} \int_0^{\pi/2} \cos^7 \phi \sin \phi \, d\phi \\ &= -\frac{\pi}{24} \left[\cos^8 \phi \right]_0^{\pi/2} \\ &= \frac{\pi}{24}. \end{aligned}$$

- Q5.** The plane through the points $(2, 0, 0)$, $(0, 2, 0)$ and $(0, 0, 2)$ has equation $x + y + z = 2$. Using this we obtain

$$T = \{(x, y, z) \mid 0 \leq x \leq 2, 0 \leq y \leq 2 - x, 0 \leq z \leq 2 - x - y\}.$$

Therefore,

$$\begin{aligned} m &= \iiint_T \rho(x, y, z) \, dV = \int_0^2 \int_0^{2-x} \int_0^{2-x-y} y^2 \, dz \, dy \, dx = \int_0^2 \int_0^{2-x} y^2(2-x-y) \, dy \, dx \\ &= \int_0^2 \left[\frac{1}{3}(2-x)y^3 - \frac{1}{4}y^4 \right]_0^{2-x} dx = \frac{1}{12} \int_0^2 (2-x)^4 \, dx = -\frac{1}{60}(2-x)^5 \Big|_0^2 = \frac{8}{15}. \end{aligned}$$

- Q6.** Let $u = y + x$ and $v = y - x$. Using the following sketch of the region R , we can see that its description in the uv -plane is $-1 \leq u \leq 1$, $-1 \leq v \leq 1$. On the other hand, $x = (u - v)/2$ and $y = (u + v)/2$ which give

$$J = \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{vmatrix} = \frac{1}{2}.$$

Now

$$\iint_R e^{y-x} \, dA = \int_{-1}^1 \int_{-1}^1 \frac{1}{2} e^v \, du \, dv = \int_{-1}^1 e^v \, dv = e^v \Big|_{-1}^1 = e - \frac{1}{e}.$$

